

# Week 2

## Introduction

Imagine sending a letter across the globe; it needs a unique address, a way to be broken down if too large for a particular mail slot, and a reliable delivery service to ensure it arrives intact. The internet operates on similar principles, but at an astonishing scale and speed. This week, we delved into the foundational mechanisms that make this possible, exploring how data packets find their way across diverse networks. We began by understanding **Lecture 06 : IP addressing and routing (Part I)**, which introduced the critical concept of IP packet fragmentation—a necessary process when data units are too large for certain network segments. This ensures data can traverse the internet's heterogeneous infrastructure.

Building on this, **Lecture 07 : IP addressing and routing (Part II)** illuminated the core of network identification: IP addresses. We uncovered their hierarchical structure, the historical class-based system, and how they uniquely identify every device. The journey then moved to the transport layer with **Lecture 08 : TCP and UDP (Part I)** and **Lecture 09 : TCP and UDP (Part II)**, where we contrasted TCP's reliable, connection-oriented service with UDP's fast, connectionless approach, and examined the crucial role of port numbers. Finally, **Lecture 10 : IP subnetting** provided insights into optimizing IP address allocation and network management through subnetting, VLSM, and CIDR. Together, these topics form the bedrock of how data is addressed, routed, and reliably delivered across the vast expanse of the internet.

## Core Concepts

| Term             | Definition                                                                                                                                                 | Significance                                                                                                                         |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| IP Fragmentation | The process of dividing a large IP packet into smaller pieces (fragments) when it encounters a network segment with a smaller Maximum Transfer Unit (MTU). | Essential for enabling data to traverse networks with varying size limitations, ensuring delivery across heterogeneous environments. |
| IP Address       | A unique 32-bit numerical label assigned to each device connected to a computer                                                                            | Ensures that data packets can be uniquely routed to                                                                                  |

| Term                   | Definition                                                                                                           | Significance                                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
|                        | network using the Internet Protocol.                                                                                 | the correct destination device across the internet.                                                                             |
| <b>Network Number</b>  | The initial portion of an IP address that identifies the specific network to which a host belongs.                   | Routers use this part to forward packets to the correct destination network, enabling efficient inter-network communication.    |
| <b>TCP</b>             | Transmission Control Protocol; a connection-oriented, reliable, full-duplex, byte-stream service protocol.           | Guarantees data integrity, order, and flow control for applications requiring high reliability, like web browsing or email.     |
| <b>UDP</b>             | User Datagram Protocol; a connectionless, unreliable datagram service protocol.                                      | Offers speed and efficiency for applications where occasional data loss is acceptable, such as live streaming or online gaming. |
| <b>Port Number</b>     | A 16-bit identifier assigned to a specific user process or application running on a computer.                        | Enables the operating system to direct incoming network packets to the correct application on a host.                           |
| <b>3-Way Handshake</b> | The three-step process TCP uses to establish a connection between a client and a server.                             | Ensures both parties are ready to communicate and synchronize initial sequence numbers before data transfer begins.             |
| <b>IP Subnetting</b>   | The process of dividing a larger IP network into smaller, distinct sub-networks, each with its own unique subnet ID. | Improves network management, enhances security, and optimizes the use of limited IP address space.                              |
| <b>VLSM</b>            | Variable Length Subnet Mask; a subnetting technique that                                                             | Maximizes IP address utilization by tailoring subnet                                                                            |

| Term        | Definition                                                                                                                               | Significance                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|             | permits the use of different subnet masks within the same IP network.                                                                    | sizes to the actual number of hosts required by different network segments.                                                                 |
| <b>CIDR</b> | Classless Inter-Domain Routing; an IP addressing method that abandons traditional Class A, B, C systems, using a flexible prefix length. | Reduces routing table sizes, improves routing efficiency, and significantly conserves IP addresses by allowing networks of arbitrary sizes. |

## Detailed Analysis

### 1. IP Packet Fragmentation and Routing

The internet is a patchwork of diverse networks, each with its own capacity for handling data units, defined by its Maximum Transfer Unit (MTU). As discussed in **Lecture 06 : IP addressing and routing (Part I)**, when a large IP packet encounters a network segment with a smaller MTU, it must be broken down into smaller, compliant fragments. This process, known as IP fragmentation, is handled by the IP layer, ensuring data can traverse heterogeneous network environments. Each fragment is then treated as an independent IP packet.

Two primary approaches to fragmentation exist:

- \* **Transparent Fragmentation:** Fragmentation and reassembly occur within the same network, making the process invisible to subsequent networks. An ingress router fragments the packet, and a designated exit router reassembles it before forwarding the original packet. While simplifying downstream handling, this can create bottlenecks at the exit router due to the reassembly burden.
- \* **Non-Transparent Fragmentation:** This is the method adopted by the IP protocol. Fragments are not reassembled by intermediate routers but are forwarded independently to the final destination host for reassembly. This allows for parallel processing and better network resource utilization but increases header overhead, as each fragment carries its own IP header.

The IP header contains specific fields crucial for non-transparent fragmentation:

- \* **Identification (ID)** : A 16-bit field that uniquely identifies the original datagram to which a fragment belongs, allowing the destination host to group related fragments.
- \* **Fragment Offset** : A 13-bit field specifying the position of the current fragment's data relative to the beginning of the original datagram, measured in multiples of 8

bytes. This helps the destination host correctly order fragments.

- \* **Flags** : A 3-bit field with two key bits: 'D' (Do Not Fragment), which prevents fragmentation and causes the packet to be dropped if too large, and 'M' (More Fragments), which indicates if more fragments are expected after the current one.

An example illustrates the overhead: an original 1020-byte packet (1000 data + 20 header) might be fragmented into four pieces, resulting in 1080 total bytes transmitted (1000 data + 4 \* 20 headers), demonstrating the increased header overhead. Practical IP MTU is often limited to 1500 bytes to align with Ethernet's maximum frame size, avoiding fragmentation at the data link layer.

## 2. IP Addressing Fundamentals

**Lecture 07 : IP addressing and routing (Part II)** introduced IP addresses as the unique 32-bit numerical labels essential for identifying devices on the internet. To make these binary addresses human-readable, dotted decimal notation is used, dividing the 32 bits into four 8-bit octets, each converted to decimal and separated by dots (e.g., 66.134.48.126 ).

IP addresses are hierarchical, comprising a network number and a host number . The network number identifies the specific network, while the host number uniquely identifies a device within that network. Routers primarily use the network number for inter-network routing, and once a packet reaches the correct network, the host number guides it to the specific device.

Historically, IP addresses were categorized into classes:

- \* **Class A**: Identified by a leading 0 . Network part is 8 bits, host part is 24 bits. Range: 0.0.0.0 to 127.255.255.255 . Supports ~16 million hosts per network.
- \* **Class B**: Identified by leading 10 . Network part is 16 bits, host part is 16 bits. Range: 128.0.0.0 to 191.255.255.255 . Supports ~65,534 hosts per network.
- \* **Class C**: Identified by leading 110 . Network part is 24 bits, host part is 8 bits. Range: 192.0.0.0 to 223.255.255.255 . Supports 254 hosts per network.
- \* **Class D**: Identified by leading 1110 . Used for multicast communication ( 224.0.0.0 to 239.255.255.255 ).
- \* **Class E**: Identified by leading 1111 . Reserved for experimental purposes ( 240.0.0.0 to 255.255.255.255 ).

Special purpose IP addresses include:

- \* **Private IP Addresses**: Ranges reserved for internal networks (e.g., 10.x.x.x , 172.16.x.x to 172.31.x.x , 192.168.x.x ), not routable on the public internet.
- \* **Loopback Address**: 127.x.x.x (e.g., 127.0.0.1 ), refers to the local machine for self-testing.
- \* **Broadcast Address**: All 1s in the host part (e.g., 255.255.255.255 ) to send a packet to all devices on a local network segment.

### 3. TCP and UDP: Transport Layer Protocols

**Lecture 08 : TCP and UDP (Part I)** introduced the two fundamental transport layer protocols: TCP (Transmission Control Protocol) and UDP (User Datagram Protocol). These protocols sit above the unreliable IP layer and provide different services to applications based on their reliability and speed requirements.

#### **TCP's Core Features:**

- \* **Connection-Oriented:** Establishes a logical connection (a "virtual circuit") before data transfer, maintained throughout the communication.
- \* **Reliable Data Transfer:** Ensures data arrives correctly, completely, and in order through mechanisms like:
- \* **Checksum:** Detects corruption.
- \* **Positive Acknowledgement:** Receiver confirms successful receipt of bytes.
- \* **Timeouts:** Sender retransmits data if no ACK is received within a set interval.
- \* **Flow Control:** Prevents sender from overwhelming receiver.
- \* **Sequencing:** Reorders out-of-order packets into the correct sequence.
- \* **Full-Duplex Communication:** Allows simultaneous two-way data transmission.
- \* **Byte-Stream Service:** Treats data as an ordered sequence of bytes, tracked by sequence numbers.

#### **UDP's Characteristics:**

- \* **Connectionless Service:** Sends each datagram independently without prior connection setup.
- \* **Unreliable Datagram Service:** Does not guarantee delivery, order, or prevent duplication; no retransmissions.
- \* **Minimal Overhead:** Simpler design leads to faster communication, ideal for latency-sensitive applications.
- \* **Checksum:** Detects corruption in UDP datagrams.
- \* **Port Numbers:** Identifies specific user processes on a host.

**Port Numbers:** Crucial for process-to-process communication, port numbers are 16-bit identifiers that direct incoming packets to the correct application on a machine.

- \* **Well-Known Port Numbers:** Standardized ports (1-1023, extended to 4095) for common services (e.g., HTTP 80, SMTP 25).
- \* **Ephemeral Port Numbers:** Temporary ports assigned dynamically to client processes for the duration of a connection.

Information on well-known ports is stored in the `services` file (e.g., `/etc/services` on Unix/Linux). A TCP connection is uniquely identified by a five-tuple "association": protocol, local IP, remote IP, local port, and remote port.

## 4. TCP and UDP: Header Formats and Connection Management

**Lecture 09 : TCP and UDP (Part II)** delved deeper into the internal mechanisms of these protocols, focusing on their header structures and TCP's connection management. TCP provides a robust, host-to-host, stream-oriented service, managing buffers and ensuring reliable, in-order delivery despite IP's unreliability. The basic unit of data transfer in TCP is a `segment`.

The **TCP Header Format** (20-60 bytes) includes:

- \* **Source/Destination Port Numbers:** Identify communicating applications.
- \* **Sequence Number:** 32-bit field indicating the byte number of the first data byte in the segment, crucial for ordering and reliability.
- \* **Acknowledgement Number:** 32-bit field indicating the next byte number the receiver expects, confirming data receipt.
- \* **Header Length:** Specifies header size in 32-bit words.
- \* **Flags:** Control bits like `SYN` (synchronize for connection setup), `ACK` (acknowledgement valid), `FIN` (no more data, for termination), `RST` (reset connection), `PSH` (push data to application), and `URG` (urgent pointer active).
- \* **Window:** 16-bit field for flow control, indicating receiver's buffer capacity.
- \* **Checksum:** For error detection across the segment and a pseudo-header.
- \* **Urgent Pointer:** Points to urgent data if the URG flag is set.
- \* **Options:** Variable field for additional features.

### **TCP Connection Establishment: The 3-Way Handshake:**

1. Client sends `SYN` (Synchronize) with an initial sequence number.
2. Server responds with `SYN-ACK` (Synchronize-Acknowledge), its own sequence number, and an ACK for the client's SYN.
3. Client sends `ACK` (Acknowledge) for the server's SYN.

This synchronizes both sides and establishes the connection.

### **TCP Connection Termination: The 4-Way Handshake:**

1. Client sends `FIN` (Finish) to close its side of the connection.
2. Server sends `ACK` for the client's FIN.
3. Server sends `FIN` when it has no more data to send.
4. Client sends `ACK` for the server's FIN, then enters a `TIME_WAIT` state before fully closing.

**Security Implications:** The 3-way handshake can be exploited in Denial-of-Service (DoS) attacks. A "half-open connection" occurs if a client sends SYN, the server responds with SYN-ACK, but the client never sends the final ACK. Malicious actors can flood a server with SYN requests, exhausting its resources and preventing legitimate connections.

The **UDP Header Format** is much simpler (fixed 8 bytes):

- \* **Source/Destination Port Numbers:** Identify applications.

- \* **Length:** Total length of UDP header and data.

- \* **Checksum:** Optional field for error detection.

UDP's simplicity results in lower overhead and faster transmission, making it suitable for applications where speed is prioritized over absolute reliability.

## 5. IP Subnetting and Address Management

**Lecture 10 : IP subnetting** explored how to efficiently manage and allocate IP addresses, moving beyond the rigid class-based system. IP subnetting divides a larger IP network into smaller, distinct sub-networks, improving organization, security, and address utilization. This is achieved using an IP mask, a 32-bit number that differentiates the network portion from the host portion of an IP address.

- **Natural Masks:** Default masks for Class A ( 255.0.0.0 ), B ( 255.255.0.0 ), and C ( 255.255.255.0 ) networks, defining their standard network/host divisions.
- **Subnetting with Flexible Masks:** Involves borrowing bits from the host portion of an IP address to create a subnet part . The subnet mask is extended to include these borrowed bits. For example, a Class A network 10.0.0.0 with a subnet mask of 255.255.0.0 borrows 8 bits, creating 256 subnets, each supporting 65534 hosts.

**Variable Length Subnet Mask (VLSM)** is an advanced technique allowing different subnet sizes within the same network. This optimizes IP address utilization by tailoring subnet sizes to actual host requirements, preventing waste. For instance, a Class C network can be divided into subnets of 128, 64, and 32 hosts to match departmental needs, rather than using fixed-size subnets that might be too large or too small.

**Classless Inter-Domain Routing (CIDR)** is the modern standard, abandoning classful addressing for a flexible prefix length ( /M ). An IP address like

144.16.192.57/18 indicates that the first 18 bits define the network. CIDR offers:

- \* **Flexible Network Sizing:** Allows arbitrary block sizes.

- \* **Reduced Routing Table Size:** Aggregates multiple smaller networks into single entries.

- \* **IP Address Conservation:** Prevents wasteful allocation.

CIDR blocks must have a total number of addresses that is a power of 2, and their starting address must be evenly divisible by the block size. All modern routers support CIDR, making it the prevalent addressing scheme today.

## Key Takeaways

- [ ] IP fragmentation is essential for data to traverse networks with varying Maximum Transfer Unit (MTU) sizes.
- [ ] Non-transparent fragmentation, used by IP, sends fragments independently to the destination for reassembly, increasing header overhead.
- [ ] IP addresses are unique 32-bit identifiers, hierarchically divided into network and host parts, and historically categorized into classes (A, B, C, D, E).
- [ ] TCP provides a reliable, connection-oriented, full-duplex service with mechanisms like sequence numbers, acknowledgements, and flow control.
- [ ] UDP offers a fast, connectionless, unreliable service, prioritizing speed and low overhead for applications where data loss is acceptable.
- [ ] Port numbers are crucial for directing network traffic to specific applications on a host, categorized as well-known or ephemeral.
- [ ] TCP connection establishment (3-way handshake) and termination (4-way handshake) ensure synchronized and graceful communication.
- [ ] Half-open TCP connections can be exploited in Denial-of-Service (DoS) attacks by exhausting server resources.
- [ ] IP subnetting, VLSM, and CIDR are techniques to efficiently manage IP address space, segment networks, and improve routing efficiency.
- [ ] CIDR, with its flexible prefix length notation ( /M ), is the modern standard for IP address allocation and routing, replacing rigid classful addressing.

## Conclusion

This week's exploration into IP addressing, fragmentation, and the transport layer protocols TCP and UDP has unveiled the intricate yet robust architecture that underpins the internet. From the necessity of breaking down large data packets to navigate diverse network capacities, to the meticulous process of assigning unique identifiers to every device, and finally, to the sophisticated mechanisms that ensure reliable or efficient data delivery, we've seen how these fundamental concepts work in concert. The evolution from classful IP addressing to the flexible paradigms of VLSM and CIDR highlights the continuous adaptation required to scale and optimize network resources in an ever-expanding digital world.

Understanding these core principles is not just academic; it's crucial for anyone interacting with or building upon network technologies. It explains why your video stream might occasionally buffer (UDP's speed over reliability) or why your online



banking transaction is secure (TCP's reliability). This foundational knowledge sets the stage for comprehending more advanced topics in network security, application development, and the future of internet infrastructure, including the transition to IPv6. The internet's ability to seamlessly connect billions of devices and deliver information globally is a testament to the ingenious design and continuous refinement of these protocols.